

Formula Sheet

Value of Test Statistic
$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; \sigma \text{ known}$
$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; v = n - 1, \sigma \text{ unknown}$
$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}; \sigma_1 \text{ and } \sigma_2 \text{ known}$
$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}; v = n_1 + n_2 - 2, \sigma_1 = \sigma_2 \text{ but unknown, } s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}; v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}, \sigma_1 \neq \sigma_2 \text{ and unknown}$
$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}; v = n - 1$

Confidence Interval estimation

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}},$$

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}},$$

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

$$\bar{d} - t_{\alpha/2} \frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2} \frac{s_d}{\sqrt{n}},$$

Simple linear regression and correlation co-efficients and test statistic for correlation co-efficient

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}.$$

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}},$$

$$v = n - 2$$

ANOVA Formulae

Sum of squares	Defining formula	Computing formula
Total, SST	$\sum (x_i - \bar{x})^2$	$\sum x_i^2 - (\sum x_i)^2/n$
Treatment, SSTR	$\sum n_j (\bar{x}_j - \bar{x})^2$	$\sum (T_j^2/n_j) - (\sum x_i)^2/n$
Error, SSE	$\sum (n_j - 1) s_j^2$	$SST - SSTR$

Correlation co-efficient and co-variance formulae for joint PMF/PDF

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y. \quad \rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

Inference on regression co-efficients ($(v = n - 2)$)

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}, \quad S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}, \quad S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n}$$

$$b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}},$$

$$b_0 - t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2} < \beta_0 < b_0 + t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2},$$

T-Statistic for β_0

$$T = \frac{\bar{B}_0 - \beta_0}{S \sqrt{\sum_{i=1}^n x_i^2 / (n S_{xx})}}$$

T-Statistic for β_1

$$T = \frac{(B_1 - \beta_1) / (\sigma / \sqrt{S_{xx}})}{S / \sigma} = \frac{B_1 - \beta_1}{S / \sqrt{S_{xx}}}$$

Total probability and Baye's Rule formulae

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) P(A|B_i). \quad P(B_r|A) = \frac{P(B_r \cap A)}{\sum_{i=1}^k P(B_i \cap A)} = \frac{P(B_r) P(A|B_r)}{\sum_{i=1}^k P(B_i) P(A|B_i)}$$

Gradient Descent Algorithm

$$b_0^{new} = b_0^{old} - \alpha \frac{\partial L}{\partial b_0}$$

$$b_1^{new} = b_1^{old} - \alpha \frac{\partial L}{\partial b_1}$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{n} \sum_1^n (b_0 + b_1 x - y) :$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{n} \sum_1^n (b_0 + b_1 x - y) x$$